Predictive Enrichment of Geometry-Induced Soft-State Abstractions: Supplementary Material

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Abstract

This supplement supports an investigation of whether reconstruction-score abstractions retain predictive distinctions and make them accessible to finite decoders. It specifies the numerical observation laws, representations, decoder construction, validation selection, and statistical analysis of the matched forecasting study. It reports every primary confidence interval and the terminal means and repetition standard deviations of all procedures under both selection rules. Numerical diagnostics distinguish mathematical information recovery from finite-precision behavior. Separate controlled and policy studies are described with their original populations and comparison families, including unfavorable findings. The reproducibility specification distinguishes document compilation, saved-loss analysis, fitted-model replay, and simulation regeneration.

Notation. Vectors are columns; $\top$ denotes transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|_F$ the Frobenius norm. The operator col concatenates column vectors, I_n is an identity matrix, and $\mathbf{1}_n$ is an all-ones vector. An unsubscripted identity has the dimension required by the expression. Expectations and conditional expectations refer to the stated probability law. All fitted transformations are fixed at evaluation time.

S1 Observation laws, data allocation, and random streams

The central study contains 64 independent repetitions of four numerical observation laws. A task is one law and repetition. Each task supplies independently generated training (A), validation (B), diagnostic (C), and terminal (D) episode sets. All compared procedures receive the same observations and retained windows in that task. There is no added measurement-noise sweep beyond the innovations and sensor noise specified below.

S1.1 Autoregression

Let u_t be the scalar observation at integer time t. In the linear law, $u_0 = u_1 = 0$ and

$$u_t = 0.65u_{t-1} - 0.15u_{t-2} + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} N(0, 0.1^2), \quad t \geq 2.$$

Discard the first 256 stored values and retain the next 72. In the nonlinear law, u_0, u_1 are independent $U[-0.8, 0.8]$ draws, and

$$u_t = \tanh\{1.25u_{t-1} - 0.65u_{t-2} + 0.25\sin(3u_{t-1}) + \epsilon_t\}, \qquad \epsilon_t \overset{\text{iid}}{\sim} U[-0.1, 0.1].$$

Discard the first 80 stored values and retain the next 72. Here $N(\mu, \sigma^2)$ denotes a normal law with mean μ and variance σ^2, and $U[a, b]$ a uniform law on $[a, b]$. Innovations are independent of initial values. Independent episodes use independent draws under the prescribed stream.

S1.2 Noisy oscillator observations

At continuous time s, position $q(s)$ and velocity $v(s)$ obey

$$\dot{q} = v, \qquad \dot{v} = (1 - q^2)v - q \qquad \text{(Van der Pol)},$$
$$\dot{q} = v, \qquad \dot{v} = -0.2v + q - q^3 + 0.3\cos(1.2s + \theta) \qquad \text{(Duffing)}.$$

Dots denote time derivatives. Initial position and velocity are independent $U[-2, 2]$ draws. An independent phase $\theta \sim U[0, 2\pi]$ is drawn for both laws but used only by Duffing. For each episode, the random-number order is initial position, initial velocity, phase, and then 72 sensor-noise values $\xi_j \overset{\text{iid}}{\sim} U[-0.02, 0.02]$.

Classical fourth-order Runge–Kutta integration uses float64 arithmetic, step 0.025, and a burn-in interval of length 10. Observations are spaced by 0.1, with the first retained observation at time 10.1. If q_j is the retained numerical position, the observed value is $u_j = 0.95\tanh(q_j) + \xi_j$. The laws are noisy numerical observation processes, not exact continuous-time trajectories or exactly stationary samples after a finite burn-in. The implementation fixes the integration and random-draw order.

S1.3 Episodes, windows, and the unit of inference

A/B/C/D contain 64/32/32/128 episodes, each of length 72. For origins $t = 11, 15, \ldots, 59$, define

$$x_t = (u_{t-11}, \ldots, u_t)^\top \in \mathbb{R}^{12}, \qquad Y_t = (u_{t+1}, u_{t+4}, u_{t+12})^\top \in \mathbb{R}^3.$$

Each episode yields thirteen windows. A/B/C uniformly sample 512/256/256 rows without replacement and sort the selected rows into episode/origin order. D retains all 1,664 rows. A later origin may use a value previously treated as an earlier target because that value is then observed; no model is refitted between origins. The evaluation is a set of rolling direct forecasts, not one autonomous trajectory forecast.

Overlapping windows, horizons, representations, and caps are not independent inferential observations. The independent unit for a law-specific comparison is a complete repetition, including its A/B-selected predictors and D aggregate loss. The four laws have separate primary results; their 256 tasks are not pooled as 256 observations for one contrast.

S1.4 Random streams and execution order

The namespace is `kahm-score-enrichment-final|2026-11-02|v1`. A seed is derived by SHA256 from the namespace, law identifier, repetition, split, and purpose. Data and row-sampling seeds use the first 16 digest bytes as a big-endian integer; the fit seed uses the first four. NumPy's `default_rng` uses these seeds. The exact 256-task roster and all streams are provided in `cohort.json`; repetitions are indexed 0 through 63. Fit randomness is derived from task identity, not worker order. Numerical library and seed specifications together define the reproducible computation.

The 2,304 top-level identifiers are distinct and differ from the 84 recorded development identifiers. This is a check of seed equality, not a proof that every realized pseudorandom sequence is disjoint; exclusions against all other historical streams were not exhaustively reconstructed. No earlier observation enters the current confidence intervals.

A fits all candidate states; B selects them. Complete fitted states and choices were sealed before C/D generation, and every C/D forecast was sealed before either set was scored. C makes no tuning or calibration decision. Target arrays are present after C/D preparation, so the seals demonstrate ordering and byte identity rather than cryptographic isolation. The protocol was fixed locally before production generation, with prior development knowledge, and is not a public preregistration.

S2 Numerical representation and decoder specification

S2.1 Reference construction and native local rank

Input coordinates are standardized using all retained A histories. Each population standard deviation at most 10^{-12} is replaced by one. Two separate k-means partitions request 2 and 4 groups, using five initializations and at most 150 iterations. Too few distinct rows reduce the requested count. An unresolved empty-group result is retried with halved group counts, with all attempts recorded. Each group supplies at most 24 histories, sampled without replacement. Reference identities are fixed before the cap sweep.

Let $Q_j \in \mathbb{R}^{n_j \times d}$ contain local reference rows, with $d = 12$, mean μ_j, and cap $c \in \{1, 2, 4, 8, 12\}$. References are converted to float32 before native PCA and stored as such; subsequent calculations use float64. The initial rank request is $\min(c, d, n_j - 1)$. Centered sample covariance uses divisor $n_j - 1$, descending eigenvalue order, and the positive-eigenvalue threshold given by float64 machine epsilon. If the smallest projected reference-coordinate range is below 10^{-3}, the requested rank is decremented and PCA is recomputed. This loop stops when the range condition holds or requested rank one is reached; a rank-one range violation is recorded, not hidden. The stored matrix P_j and its actual column count determine the effective rank.

There is no additional ambient-minus-one restriction. Singleton or constant reference sets use an explicitly labeled rank-zero wrapper: all reconstruction rules return μ_j and there is no local-coordinate block. This is not a successful positive-rank native fit. Unexpected factorization errors stop execution. Full-rank affine projection returns the input exactly in the implementation, avoiding numerical angular artifacts around its mathematical identity.

S2.2 Reconstruction, reference regularization, and scoring

For positive rank r_j, define S_j as the sample covariance of $P_j^\top q_{ji}$ and use the kernel

$$\kappa_j(z, z') = \exp\left\{-\frac{[P_j^\top(z - z')]^\top S_j^{-1}[P_j^\top(z - z')]}{2r_j}\right\}.$$

Let G_j contain reference kernel values and $k_j(z)$ query/reference values. For one model suppress j, set $n = n_j$, and define reference energy $\nu = \|Q\|_F^2/(nd)$. Initialize $\lambda^{(0)} = 2.5\nu$. At iteration a, compute

$$C^{(a)} = (G + \lambda^{(a)} I_n)^{-1}, \qquad \lambda^{(a+1)} = 2\nu + \frac{\|Q - GC^{(a)}Q\|_F^2}{nd}.$$

Stop when the absolute change is at most 10^{-3} or after 100 inverse computations. Use the last computed inverse and its corresponding regularizer, not the uncomputed subsequent proposal. A nonpositive or nonfinite positive-rank regularizer is an error. This fits reference reconstruction without forecast targets.

The three maps are $Q^\top Ck/(\mathbf{1}^\top Ck)$ for KAHM, $Q^\top k/(\mathbf{1}^\top k)$ for positive-kernel reconstruction, and $\mu + PP^\top(z - \mu)$ for affine projection. Query squared distances use explicit differences after the projected metric transform, rather than subtraction of large expanded squared norms. Subtracting the largest log-kernel entry before exponentiation rescales a row by a common positive factor and leaves its exact normalized quotient unchanged.

For signed weights $a = Ck$, the relative denominator is $|\mathbf{1}^\top a|/\|a\|_1$ when the denominator is positive. The threshold is 10^{-10}; a negative sum alone is not a failure. A failed finite-value or denominator check uses affine reconstruction and then recomputes the complete radial-and-angular score. No invalid query is assigned score zero merely because it used fallback.

For reconstruction $\widehat{z}$, let $e = 1 - \exp(-\|z - \widehat{z}\|_2)$ and θ be the clipped cosine angle divided by π. A zero norm sets the cosine to zero. The score is $T = \sqrt{(e^2 + \theta^2)/2}$. With $\tau = 10^{-6}$ and $\omega = 16$, each partition forms log affinities $a_j = \omega \log(1 - T_j + \tau)$, log normalizer $\zeta = \log \sum_j \exp(a_j)$, and memberships $\phi_j = \exp(a_j - \zeta)$. Log-sum-exp evaluates ζ. There is no small-sum override to uniform memberships. Score clipping to $[0, 1]$ protects against rounding; the inverse relation recovers the clipped scores. The origin convention gives the common score $1/(2\sqrt{2})$ for exact zero reconstruction.

S2.3 Feature transformation and the nonlinear decoder

The four input maps are ϕ, (ϕ, ζ), $\mathrm{col}(\phi, \{\phi_j P_j^\top (z - \mu_j)\}_j)$, and T. Each map uses its own reconstruction scores. All column means and population standard deviations are fitted on A. Standard deviations below 10^{-8} are replaced by one. The membership scaling is shared exactly across its enriched maps at the same cap, and an appended block receives its own column scaling. There is no deletion, radial transformation, or hidden raw-input augmentation.

Let $u \in \mathbb{R}^p$ be a standardized encoding. Linear ridge uses u directly. The nonlinear decoder selects 64 A rows as landmarks with identical row identities across all representations and caps. Let δ be the median squared distance between distinct landmark positions, retaining distances greater than 10^{-14}. If there are no such distances, set $\delta = 1$. For multiplier $b \in \{0.25, 0.5, 1, 2, 4\}$, set $\sigma^2 = \max(\delta b^2, 10^{-12})$.

Define the 64-by-64 Gaussian landmark Gram matrix K_{LL} and symmetrize it before eigendecomposition. Retain eigenvalues $\lambda_i > \max(10^{-12}, 10^{-10}\lambda_{\max})$, with corresponding eigenvectors in V. For the column vector $k_L(u)$ of landmark similarities and the diagonal retained-eigenvalue matrix Λ, the raw Nyström vector is $\Lambda^{-1/2} V^\top k_L(u)$. Standardize only this new block with its A moments, and append it to the unchanged linear vector u. The result is a finite Nyström decoder with a linear component (Williams & Seeger, 2001). Its width and retained kernel rank may vary between representations despite the same landmark budget.

S2.4 Ridge fitting and B selection

For A feature matrix X_f with n_A rows and three-column target matrix Y, the objective is mean row-wise squared error plus $\alpha \|W\|_F^2$ with an unpenalized intercept. Let X_c, Y_c be centered matrices, and let $X_c = U \Sigma V^\top$ be a singular value decomposition (SVD). Each ridge candidate uses

$$\widehat{W} = V \,\mathrm{diag}\left(\frac{s_i}{s_i^2 + n_A \alpha}\right) U^\top Y_c,$$

where s_i are singular values. The intercept is the target mean minus the feature mean multiplied by $\widehat{W}$. Penalties are $10^{-6}, 10^{-5}, \ldots, 10^{2}$. The same penalty and nonlinear bandwidth are selected across the three horizons; the fitted coefficient columns are distinct. A summed-squared-error implementation uses penalty $n_A \alpha$.

For each candidate, B computes the mean of its horizon MSEs divided by the corresponding retained-A target variances, floored at 10^{-12}. The first exact minimum in ascending cap, bandwidth, and penalty order is selected. Raw input has no duplicated cap search. The primary procedure independently selects the cap for each reconstruction/encoding/decoder combination. The secondary common cap is selected by affine-local linear ridge; each procedure retains its own best within-cap B settings. No C target affects a choice, and there is no A+B refit.

The menu contains 26 complete procedures, 366 candidate decoder bases, and 3,294 ridge heads per task. Counts of heads are search costs, not independent experimental units. The grid was expanded uniformly after development boundary selections, before the final data were generated. The 64-landmark budget is fixed, not selected; this study does not test a landmark-capacity interaction.

S2.5 Numerical recovery and failure accounting

Only a NumPy SVD `LinAlgError` invokes the specified SciPy `gesvd` recovery. It uses the same centered matrices and objective and checks reconstruction, orthogonality, and normal-equation residuals. It does not change penalties, add jitter, remove columns, or delete a candidate. Other unresolved failures stop the stage. Predictions must be finite, and a complete-cohort audit is required for inferential release.

The independent coefficient audit first uses Cholesky normal equations. If factorization or agreement fails the declared tolerances, it solves the same problem by augmented QR using SciPy's `gelsy` driver:

$$\begin{bmatrix} X_c \\ \sqrt{n_A \alpha}\, I \end{bmatrix} W \;\approx\; \begin{bmatrix} Y_c \\ 0 \end{bmatrix}.$$

The augmented design must have full column rank, and its coefficients and forecasts must agree under the unchanged tolerances. Every QR use is recorded. Coefficient and forecast comparisons use componentwise relative tolerance 2×10^{-7} and absolute tolerance 2×10^{-9}. There is no complete-case analysis, substitute seed, reduced comparison family, or outcome-dependent grid extension.

S3 Complete statistical results

S3.1 Loss, comparison family, and precision

For one law ℓ, repetition r, and procedure m, define

$$L_{\ell rm} = \frac{1}{3} \sum_{h \in \{1,4,12\}} \frac{1}{1664} \sum_{i \in D} \frac{(y_{ih} - \widehat{y}_{mih})^2}{v_{\ell rh}},$$

where $v_{\ell rh}$ is the retained-A horizon-target population variance floored at 10^{-12}. Each horizon and each D origin receives equal weight. The random A normalization is part of the procedure's estimand. No terminal scale fitting, trimming, winsorization, or retrospective horizon reweighting is used.

For candidate a and reference b, define $d_r = L_{\ell ra} - L_{\ell rb}$, its mean $\bar{d}$, and sample standard deviation s_d over 64 repetitions. All 40 comparisons use

$$[\bar{d} - t_{63,1-0.05/80} s_d/8,\ \bar{d} + t_{63,1-0.05/80} s_d/8], \qquad t_{63,0.999375} = 3.37962646388.$$

The marginal confidence coefficient is 99.875%; Bonferroni targets 95% simultaneous coverage when the paired-Student marginal approximation is valid. Dependence among comparisons does not invalidate the Bonferroni inequality, but nonnormal repetition differences can affect the marginal approximation. A zero sample variance would be flagged as no Student inference while retaining its family slot. None of the 40 released differences has that exception.

The fixed precision rule selected the first repetition count in $32, 48, 64, 96$ giving $t_{n-1,1-0.05/80}/\sqrt{n} \leq 0.45$. The resulting multipliers are 0.627711, 0.495723, 0.422453, and 0.339547. This is a criterion relative to a future paired standard deviation, not an absolute-error tolerance or a power guarantee. No optional stopping or sample-size extension followed terminal scoring.

There are no practical superiority, noninferiority, or equivalence margins in this family. A negative interval supports lower candidate mean loss; a positive interval supports higher mean loss; an interval containing zero is unresolved. Relative gain is $100(1 - \overline{L}_a/\overline{L}_b)$, not an average of per-repetition percentages. Confidence endpoints are absolute SMSE differences, not percentage bounds.

S3.2 All primary intervals

Tables S1–S4 contain the full primary family. Each local comparison uses KAHM-local as candidate; the normalizer comparison uses KAHM-normalizer. Membership references are KAHM-memberships. Positive-kernel references use matched positive-kernel-local features, not a different decoder. Every row uses independent B selection.

Table S1: Linear autoregression: all ten primary comparisons. Gain is positive when the candidate mean is lower. The adjusted interval and $\bar{d}$ use absolute SMSE units.

Decoder	KAHM candidate / reference	Gain (%)	$\bar{d}$	Adjusted interval	Direction
Linear	Local / memberships	+6.756	-0.0674456	[-0.0726191, -0.0622721]	Lower
Linear	Normalizer / memberships	-0.010	9.742e-05	[-0.00100141, 0.00119625]	Unresolved
Linear	Local / affine-local	-1.908	0.0174294	[0.0142194, 0.0206395]	Higher
Linear	Local / positive-kernel	+0.289	-0.00270015	[-0.00506286, -0.000337434]	Lower
Linear	Local / raw	-1.970	0.0179786	[0.0146274, 0.0213299]	Higher
Nonlinear	Local / memberships	+6.772	-0.0678589	[-0.0730383, -0.0626796]	Lower
Nonlinear	Normalizer / memberships	-0.217	0.00217696	[-0.000801873, 0.00515579]	Unresolved
Nonlinear	Local / affine-local	-1.813	0.0166392	[0.0127722, 0.0205062]	Higher
Nonlinear	Local / positive-kernel	+0.153	-0.00142702	[-0.00402884, 0.0011748]	Unresolved
Nonlinear	Local / raw	-0.427	0.00397678	[0.000308057, 0.0076455]	Higher

Table S2: Nonlinear autoregression: all ten primary comparisons. All directions remain unresolved under the specified interval calculation.

Decoder	KAHM candidate / reference	Gain (%)	$\bar{d}$	Adjusted interval	Direction
Linear	Local / memberships	+10.288	-0.00238614	[-0.00544156, 0.000669276]	Unresolved
Linear	Normalizer / memberships	-0.113	2.63198e-05	[-5.41591e-05, 0.000106799]	Unresolved
Linear	Local / affine-local	-4.417	0.000880274	[-0.00211658, 0.00387713]	Unresolved
Linear	Local / positive-kernel	-1.295	0.000266007	[-0.000776643, 0.00130866]	Unresolved
Linear	Local / raw	+5.242	-0.0011511	[-0.00421236, 0.00191016]	Unresolved
Nonlinear	Local / memberships	+6.266	-0.00146152	[-0.00539605, 0.00247301]	Unresolved
Nonlinear	Normalizer / memberships	-146.047	0.0340667	[-0.0519456, 0.120079]	Unresolved
Nonlinear	Local / affine-local	-6.756	0.0013836	[-0.00252123, 0.00528843]	Unresolved
Nonlinear	Local / positive-kernel	-5.671	0.00117346	[-0.00275127, 0.00509818]	Unresolved
Nonlinear	Local / raw	-8.444	0.00170251	[-0.00222525, 0.00563027]	Unresolved

Table S3: Van der Pol: all ten primary comparisons. Favorable raw-input results do not imply superiority over every reconstruction control.

Decoder	KAHM candidate / reference	Gain (%)	$\bar{d}$	Adjusted interval	Direction
Linear	Local / memberships	+83.634	-0.254332	[-0.266058, -0.242605]	Lower
Linear	Normalizer / memberships	+6.369	-0.0193669	[-0.0284404, -0.0102935]	Lower
Linear	Local / affine-local	-22.947	0.00928905	[0.0071626, 0.0114155]	Higher
Linear	Local / positive-kernel	+12.790	-0.00729898	[-0.0092407, -0.00535726]	Lower
Linear	Local / raw	+33.831	-0.0254465	[-0.0276059, -0.023287]	Lower
Nonlinear	Local / memberships	+99.284	-0.29743	[-0.456767, -0.138093]	Lower
Nonlinear	Normalizer / memberships	+78.075	-0.233894	[-0.393605, -0.0741823]	Lower
Nonlinear	Local / affine-local	+2.508	-5.52027e-05	[-0.000103819, -6.58674e-06]	Lower
Nonlinear	Local / positive-kernel	+0.038	-8.13202e-07	[-3.72034e-05, 3.5577e-05]	Unresolved
Nonlinear	Local / raw	+62.093	-0.00351461	[-0.00370048, -0.00332874]	Lower

Table S4: Duffing: all ten primary comparisons, including the adverse nonlinear-decoder outcomes.

Decoder	KAHM candidate / reference	Gain (%)	$\bar{d}$	Adjusted interval	Direction
Linear	Local / memberships	+91.621	-0.201463	[-0.213137, -0.189789]	Lower
Linear	Normalizer / memberships	+5.569	-0.0122454	[-0.0193366, -0.00515425]	Lower
Linear	Local / affine-local	+1.040	-0.000193624	[-0.000664596, 0.000277348]	Unresolved
Linear	Local / positive-kernel	+1.917	-0.000360041	[-0.000713809, -6.27249e-06]	Lower
Linear	Local / raw	+4.693	-0.000907296	[-0.00123192, -0.000582669]	Lower
Nonlinear	Local / memberships	+95.607	-0.294388	[-0.527117, -0.0616589]	Lower
Nonlinear	Normalizer / memberships	+58.386	-0.17978	[-0.411645, 0.0520854]	Unresolved
Nonlinear	Local / affine-local	-20.167	0.00227002	[0.0014839, 0.00305614]	Higher
Nonlinear	Local / positive-kernel	-2.490	0.000328629	[-0.000336698, 0.000993956]	Unresolved
Nonlinear	Local / raw	-22.961	0.00252578	[0.00197647, 0.00307509]	Higher

S3.3 Every procedure under both selection rules

Tables S5–S8 report terminal mean SMSE with the sample standard deviation across 64 repetitions in parentheses. Standard deviations describe variation; they are not confidence intervals. Each table includes all three reconstructions and four encodings, together with raw input. The shared-cap tables describe a different procedure and carry no additional inferential family.

Table S5: Independent selection, linear decoder: mean SMSE (repetition standard deviation).

Representation	Linear AR	Nonlinear AR	Van der Pol	Duffing
Affine local	0.913372 (0.0513)	0.0199275 (0.00195)	0.0404811 (0.00299)	0.0186173 (0.0015)
Affine memberships	0.998312 (0.0551)	0.0247294 (0.00965)	0.259083 (0.0345)	0.192394 (0.0508)
Affine normalizers	0.999896 (0.0561)	0.0252464 (0.011)	0.253046 (0.035)	0.174993 (0.0499)
Affine scores	0.993545 (0.0554)	0.0271702 (0.00246)	0.186576 (0.0254)	0.117988 (0.0449)
KAHM local	0.930801 (0.0523)	0.0208078 (0.00742)	0.0497702 (0.0052)	0.0184237 (0.00143)
KAHM memberships	0.998247 (0.056)	0.023194 (0.00237)	0.304102 (0.031)	0.219887 (0.0281)
KAHM normalizers	0.998344 (0.0561)	0.0232203 (0.0024)	0.284735 (0.0302)	0.207641 (0.0325)
KAHM scores	0.99035 (0.0557)	0.0261199 (0.00208)	0.170774 (0.0242)	0.137161 (0.0374)
Positive-kernel local	0.933502 (0.0538)	0.0205418 (0.0051)	0.0570692 (0.00396)	0.0187838 (0.00159)
Positive-kernel memberships	0.99847 (0.0558)	0.0232283 (0.00236)	0.363625 (0.0258)	0.230041 (0.0229)
Positive-kernel normalizers	0.998116 (0.0567)	0.023218 (0.00237)	0.326763 (0.0364)	0.218384 (0.0272)
Positive-kernel scores	0.983515 (0.057)	0.0279912 (0.00217)	0.178156 (0.0307)	0.136044 (0.0308)
Raw history	0.912823 (0.0515)	0.0219589 (0.00161)	0.0752167 (0.00207)	0.019331 (0.0014)

Table S6: Independent selection, nonlinear decoder: mean SMSE (repetition standard deviation).

Representation	Linear AR	Nonlinear AR	Van der Pol	Duffing
Affine local	0.917588 (0.0517)	0.0204806 (0.00184)	0.00220087 (0.000177)	0.0112563 (0.00161)
Affine memberships	1.00333 (0.0558)	0.0303263 (0.023)	0.0829511 (0.0332)	0.108187 (0.0443)
Affine normalizers	1.00531 (0.0567)	0.0473456 (0.133)	0.0606483 (0.0268)	0.0722325 (0.0546)
Affine scores	0.998349 (0.0558)	0.022348 (0.00237)	0.00301498 (0.000971)	0.025157 (0.0062)
KAHM local	0.934227 (0.0533)	0.0218642 (0.00949)	0.00214567 (0.000177)	0.0135263 (0.00167)
KAHM memberships	1.00209 (0.056)	0.0233257 (0.00236)	0.299576 (0.377)	0.307914 (0.551)
KAHM normalizers	1.00426 (0.0564)	0.0573924 (0.204)	0.0656819 (0.0259)	0.128135 (0.0303)
KAHM scores	0.997846 (0.0564)	0.0228505 (0.00339)	0.00266114 (0.000405)	0.022652 (0.0041)
Positive-kernel local	0.935654 (0.0529)	0.0206908 (0.00232)	0.00214648 (0.00016)	0.0131977 (0.0015)
Positive-kernel memberships	1.00452 (0.0551)	0.0234312 (0.00232)	0.287731 (0.0221)	0.250766 (0.116)
Positive-kernel normalizers	1.00289 (0.0554)	0.0295164 (0.0178)	0.0359837 (0.0184)	0.124334 (0.03)
Positive-kernel scores	0.992957 (0.0562)	0.0223366 (0.00203)	0.00237783 (0.00028)	0.0189918 (0.00312)
Raw history	0.93025 (0.052)	0.0201617 (0.00166)	0.00566028 (0.000414)	0.0110005 (0.00103)

Table S7: Shared cap, linear decoder: mean SMSE (repetition standard deviation). The bank is selected by affine-local linear ridge on B.

Representation	Linear AR	Nonlinear AR	Van der Pol	Duffing
Affine local	0.913372 (0.0513)	0.0199275 (0.00195)	0.0404811 (0.00299)	0.0186173 (0.0015)
Affine memberships	1.0073 (0.0567)	0.0397153 (0.13)	0.282593 (0.0377)	0.327854 (0.286)
Affine normalizers	1.00732 (0.0567)	0.0399373 (0.13)	0.274762 (0.0371)	0.318374 (0.29)
Affine scores	1.00702 (0.0568)	0.0436278 (0.129)	0.230842 (0.046)	0.300497 (0.297)
KAHM local	0.930936 (0.052)	0.0200233 (0.00221)	0.0515293 (0.00589)	0.0243328 (0.0106)
KAHM memberships	0.998938 (0.0562)	0.023234 (0.00236)	0.317966 (0.028)	0.24231 (0.0192)
KAHM normalizers	0.999011 (0.0563)	0.0232617 (0.00237)	0.301501 (0.0285)	0.237354 (0.0235)
KAHM scores	0.992703 (0.0561)	0.0262844 (0.00216)	0.203777 (0.0276)	0.173194 (0.039)
Positive-kernel local	0.93262 (0.0525)	0.0200202 (0.00208)	0.0597589 (0.00442)	0.0245053 (0.0106)
Positive-kernel memberships	0.998994 (0.056)	0.0232469 (0.00235)	0.384141 (0.0341)	0.246434 (0.0188)
Positive-kernel normalizers	0.99856 (0.0565)	0.0232517 (0.00236)	0.353534 (0.0432)	0.242304 (0.0195)
Positive-kernel scores	0.983685 (0.0575)	0.0284524 (0.0023)	0.215645 (0.0359)	0.186963 (0.0298)
Raw history	0.912823 (0.0515)	0.0219589 (0.00161)	0.0752167 (0.00207)	0.019331 (0.0014)

Table S8: Shared cap, nonlinear decoder: mean SMSE (repetition standard deviation). Each procedure selects its own within-cap decoder settings.

Representation	Linear AR	Nonlinear AR	Van der Pol	Duffing
Affine local	0.917831 (0.0517)	0.0206728 (0.00196)	0.0175849 (0.00955)	0.0138209 (0.00311)
Affine memberships	1.00738 (0.0566)	0.0455446 (0.133)	0.0983219 (0.0386)	0.29377 (0.304)
Affine normalizers	1.00737 (0.0566)	0.0422198 (0.129)	0.135489 (0.0403)	0.291139 (0.311)
Affine scores	1.00712 (0.0569)	0.039608 (0.13)	0.0339719 (0.0126)	0.291984 (0.333)
KAHM local	0.933678 (0.0529)	0.021082 (0.00304)	0.0185972 (0.012)	0.0150693 (0.00266)
KAHM memberships	1.00343 (0.0568)	0.0234263 (0.00237)	0.271928 (0.0382)	0.244937 (0.0251)
KAHM normalizers	1.00348 (0.0566)	0.0627502 (0.206)	0.107425 (0.038)	0.197443 (0.0427)
KAHM scores	0.999656 (0.056)	0.0229556 (0.00262)	0.00964096 (0.00347)	0.063873 (0.0361)
Positive-kernel local	0.935745 (0.0526)	0.0209577 (0.00253)	0.0190622 (0.0137)	0.0151556 (0.00321)
Positive-kernel memberships	1.00421 (0.0565)	0.0234538 (0.00236)	0.298582 (0.0256)	0.248453 (0.0177)
Positive-kernel normalizers	1.00321 (0.0562)	0.0320584 (0.0241)	0.06329 (0.0309)	0.17598 (0.0307)
Positive-kernel scores	0.994421 (0.0565)	0.0225586 (0.00225)	0.00549268 (0.00172)	0.0402177 (0.0161)
Raw history	0.93025 (0.052)	0.0201617 (0.00166)	0.00566028 (0.000414)	0.0110005 (0.00103)

S3.4 Descriptive comparisons and sensitivity

Five of the 40 point orderings change sign between the independent and shared-cap views. Table S9 shows the KAHM/positive-kernel comparisons in both views. The common cap is chosen for affine-local linear ridge, not independently for either of these nonlinear reconstructions. Its results cannot replace primary decisions.

For a fixed reconstruction and common cap, memberships plus normalizers and full scores contain the same information in exact arithmetic. Table S10 reports the difference between their fitted mean losses for every law, decoder, and reconstruction. The conditioning, scaling, kernel, and penalty are parts of the finite procedure. Thus these differences do not measure information gained or lost. Independently selected caps can change the underlying score maps and do not instantiate the same information-equivalent pair.

Table S9: KAHM-local versus positive-kernel-local: point relative gains under both cap rules. Shared-cap values are descriptive. Only the last column reports a primary interval decision.

Law	Decoder	Independent gain (%)	Shared gain (%)	Primary direction
Linear AR	Linear	+0.289	+0.181	Lower
Linear AR	Nonlinear	+0.153	+0.221	Unresolved
Nonlinear AR	Linear	-1.295	-0.015	Unresolved
Nonlinear AR	Nonlinear	-5.671	-0.593	Unresolved
Van der Pol	Linear	+12.790	+13.771	Lower
Van der Pol	Nonlinear	+0.038	+2.440	Unresolved
Duffing	Linear	+1.917	+0.704	Lower
Duffing	Nonlinear	-2.490	+0.570	Unresolved

Table S10: Same-bank normalizer-encoding loss minus full-score loss, in mean SMSE. Negative values favor normalizers. All 24 values are descriptive, without new significance labels.

Law	Decoder	KAHM	Positive-kernel	Affine
Linear AR	Linear	0.00630865	0.0148753	0.000307013
Linear AR	Nonlinear	0.00382131	0.00878512	0.000249294
Nonlinear AR	Linear	-0.00302266	-0.00520071	-0.0036905
Nonlinear AR	Nonlinear	0.0397946	0.00949972	0.00261188
Van der Pol	Linear	0.0977242	0.137889	0.0439203
Van der Pol	Nonlinear	0.0977839	0.0577974	0.101517
Duffing	Linear	0.0641606	0.0553412	0.0178772
Duffing	Nonlinear	0.13357	0.135763	-0.000844685

The primary numerical record includes all 2,560 paired differences, as well as each sequence's median, extrema, favorable count, and leave-one-out mean range. For nonlinear autoregression with the nonlinear decoder, normalizer enrichment is worse in 63 repetitions but its mean-difference interval is unresolved; local enrichment is better in 61 but its interval is also unresolved. Extreme observations remain included. The primary mean estimand is not replaced by the fraction of favorable repetitions, and no post hoc normality pretest selects an alternative inferential method.

The machine-readable release also contains every C diagnostic mean, D horizon decomposition, within-cap score, and cap/penalty/bandwidth frequency. These are descriptive analyses using the same cohort, not additional independent tests. Final-grid boundary selections remain limitations of the prescribed procedure rather than permission to retune after observing D. In particular, the single 64-landmark design cannot support a claim about a landmark-capacity interaction.

S4 Numerical diagnostics, costs, and verification

The completed cohort contains 7,680 local-model/cap records. All have positive rank; no rank-zero wrapper or projected-range reduction was used. Native reference constructions were compared against separate calls to the pinned autoencoder on the same reference arrays. The maximum recorded differences in bases, projected references, regularized inverses, and square-root metrics were zero. These are parity checks using shared primitives, not independent implementations of native PCA or KAHM.

The stack of local basis transposes has numerical column rank 12 in 254 of 256 cap-two banks and all cap-four, cap-eight, and cap-twelve banks. No cap-one stack has full column rank. The maximum recorded projected-metric condition number is approximately 2.05×10^5. The smallest observed memberships are 1.90×10^{-44} for KAHM, 7.53×10^{-46} for positive-kernel reconstruction, and 3.35×10^{-43} for affine projection. These sampled quantities are not uniform bounds over an unobserved input domain.

The largest membership partition-sum discrepancy is 3.55×10^{-15}, and the largest numerical error in recovering scores from memberships and normalizers is 2.78×10^{-16}. The minimum observed signed relative denominator is 1.88×10^{-5}. There were no KAHM or positive-kernel query fallbacks. Nevertheless, 309,327 repeated local-model/query evaluations would have had a completely zero kernel row without log-kernel rescaling. The counts refer to evaluations across models, caps, and splits, not independent observations. Constructed tests separately cover cancellation, negative denominators, rank-zero data, and affine fallback with recomputed scores.

S4.1 Whole-search resources

Table S11 reports whole-search measurements. Task wall and CPU values are sums across workers, whereas elapsed stage time includes orchestration and is not their sum. Peak resident set size (RSS) is the largest per-worker observation, not aggregate simultaneous memory. The same computations contribute to several procedures, so these totals are not isolated KAHM or RBF deployment costs. Stored fits include all candidates

rather than a minimal selected model. Audit, reporting, compression, and document builds are not included in fit or prediction costs.

Table S11: Recorded resources for the complete 256-task search. MiB denotes 2^{20} bytes.

Stage	Task wall (s)	Task CPU (s)	Max. RSS (MiB)	Elapsed (s)
prepare-ab	13.24444	13.21293	162.617	175.5729
fit	965.2781	962.8252	303.754	490.7443
prepare-cd	16.83628	16.75609	195.754	190.7447
predict	411.7569	410.627	271.918	295.7303

S4.2 Computational audit

The released audit covers 256 tasks, 843,264 candidate ridge-head refits, 62,464 C/D forecast-array replays, 688,128 reconstructed windows, and 26,624 selected loss records. All 40 primary intervals and decisions were independently recomputed from the saved losses. The largest absolute discrepancy in B forecast calculations is 3.84×10^{-8} under the componentwise tolerances stated in Section S2. The maximum recorded discrepancies in C/D replay and selected-loss reconstruction are zero. A nonzero absolute discrepancy can be acceptable when it satisfies the combined relative and absolute tolerance at that element.

Production used one checked SVD recovery. The independent audit used its prespecified augmented-QR calculation for 248 candidate heads. Neither changes the predictor or its tuning menu. All candidates and tasks were retained. The original complete audit, source manifest, stage identities, and final release receipt are provided; the audit was not inferred from successful document compilation.

The audit shares the feature encoder and native routines. It reconstructs windows from saved episodes and uses distinct ridge calculations, but it does not independently reimplement the physical integrators or partition fitting. It is internal computational verification, not replication by an independent group. Re-executing the recorded seeds would provide computational reproduction, not additional independent repetitions.

S5 Related controlled and forecasting evaluations

The following studies address related questions with different populations, rank policies, encodings, or readouts. They informed the central design but contribute no observations to its 64-repetition confidence family. Complete records remain in study-specific research archives. A favorable result in one procedure does not repair an unfavorable result in another.

S5.1 A target intervention with fixed inputs and geometry

Let S take -1 and 1 with equal probability, let $T \sim U[-1, 1]$, and let $\epsilon \sim U[-0.1, 0.1]$, independently. For curvature $c \in \{0, 1\}$ and mixture weight $\gamma \in \{0, 1/2, 1\}$, set

$$X = (T,\ c\cos(\pi T) + 0.5S)^\top, \qquad Y_\gamma = \sqrt{1-\gamma}\, S + \sqrt{3\gamma}\, T + \epsilon.$$

Within a repetition and curvature, changing γ leaves the inputs, noise values, and fitted geometry unchanged. The predictable signal has variance one and the observation-noise variance is $1/300$. The target changes from branch identity to signed position without changing those variances.

For straight branches ($c = 0$), the recorded affine reconstruction is almost exact, but its score memberships nearly preserve the reflection $(x_1, x_2) \mapsto (-x_1, x_2)$. A representation exactly invariant to this reflection cannot encode the sign of T. Symmetry gives $\mathbb{E}[T \mid \Gamma(X)] = 0$; hence a position component with variance γ is lost in squared-error prediction even when branch identity is retained. The fitted numerical representation is only approximately invariant, so its measured reflection discrepancy is a diagnostic rather than a population certificate.

The complete six branch contrasts in that study compare affine and KAHM memberships. With $c = 0$ and $\gamma = 0$, their reported MSEs are 0.003334 and 0.003347; at $\gamma = 1/2$ they are 0.503448 and 0.013995; at $\gamma = 1$ they are 1.002694 and 0.024668. With $c = 1$, the corresponding pairs are 0.282887/0.281126, 0.160813/0.165207, and 0.062366/0.051042. A separate locality-aware affine score gives 0.004641 in the straight-branch position case, showing that useful position information is not exclusive to KAHM. Local-coordinate controls also restore predictive access. The complete original 29-comparison family, including forecasting and static-regression controls, is preserved in the predictive-geometry archive; its intervals are not reinterpreted as members of the central family.

S5.2 Finite decoder capacity with constrained local ranks

A separate follow-up uses 24 fresh repetitions for each of three nonlinear laws, 64- and 256-landmark nonlinear decoders, and score-based local features. Its local rank is constrained by $\min(2, d-1, n_j-1)$, unlike the validation-selected native dimensions of the central study. The reference partitions request 2, 4, and 8 groups. Its normalization, query-underflow handling, and fitted-head conventions also differ; a same-sample stabilization analysis is descriptive rather than a new independent cohort.

Table S12 retains all twelve primary capacity comparisons. The difference is control MSE minus KAHM MSE, so positive values favor KAHM, the opposite sign from the central candidate-minus-reference convention. Each interval is $\bar{D} \pm t_{23,1-0.05/(2\cdot12)} s_D/\sqrt{24}$, with critical value 3.18066309590. A practical benefit requires its lower endpoint to exceed the study's fixed positive margin. An unfavorable interval lies below zero. The family has four practical-benefit results, six unfavorable comparisons, and two unresolved signs. The margins and population are specific to that study; they are not used in the central experiment.

At 256 landmarks on Van der Pol, raw, affine-local, and KAHM-local mean MSEs are 0.001654150, 0.001078003, and 0.000871969. KAHM's point gains are 47.3% and 19.1%. Both raw-input comparisons on nonlinear autoregression favor raw inputs, and all four Duffing control comparisons favor the controls. Thus the favorable Van der Pol results coexist with adverse outcomes in the same complete family. Their different capacities do not by themselves constitute a primary statistical interaction test.

Table S12: Complete 24-repetition capacity family. Improvement, interval endpoints, and margins are multiplied by 10^3. Positive improvement favors KAHM-local. These are MSEs from a different procedure, not the central SMSE estimand.

Law	Landmarks	Control	Improvement	Adjusted interval	Margin	Conclusion
Nonlinear AR	64	Raw	-0.45564	[-0.52353, -0.38775]	0.22012	Higher loss
Nonlinear AR	64	Affine local	0.03249	[-0.0030589, 0.06804]	0.22012	Unresolved
Nonlinear AR	256	Raw	-0.47912	[-0.60133, -0.3569]	0.22012	Higher loss
Nonlinear AR	256	Affine local	0.05983	[-0.013067, 0.13273]	0.22012	Unresolved
Van der Pol	64	Raw	1.832	[1.7369, 1.9271]	0.14891	Margin met
Van der Pol	64	Affine local	0.3501	[0.27552, 0.42468]	0.14891	Margin met
Van der Pol	256	Raw	0.78218	[0.75496, 0.8094]	0.14891	Margin met
Van der Pol	256	Affine local	0.20603	[0.181, 0.23107]	0.14891	Margin met
Duffing	64	Raw	-0.75566	[-1.0831, -0.42818]	0.17269	Higher loss
Duffing	64	Affine local	-1.1605	[-1.4707, -0.85028]	0.17269	Higher loss
Duffing	256	Raw	-1.2372	[-1.5242, -0.95025]	0.17269	Higher loss
Duffing	256	Affine local	-1.2756	[-1.5702, -0.98101]	0.17269	Higher loss

S5.3 A different gate-and-departure experiment

A four-arm representation study uses center-distance gates G, adds gate-weighted local coordinates to form P, and appends KAHM or matched positive-kernel reconstruction departures to form K or R. It does not use the score memberships studied here. Its 32 independent repetitions yield 256 law/noise tasks. The primary bank is selected by P, and all four arms use direct affine ridge heads. Its one-sided four-comparison family asks for 3% nonlinear superiority or 5% linear noninferiority.

The nonlinear P/G gain is 69.201%, the nonlinear K/P gain 10.459%, and the nonlinear K/R gain 0.029%. The first two specified margins are established; the third is not. Linear K loss is 1.111% higher than P while satisfying the 5% tolerance. The failed K/R margin is not equivalence. These results isolate local position and departure enrichment under their chosen gate map. They are not a direct confirmation of the reconstruction-score hierarchy or an autonomous propagation test.

S5.4 Residual, policy, and natural-series evaluations

A validation-selected dimension study uses 24 repetition units and 192 law/noise tasks, retains a raw-history ridge forecast, and fits geometric features to residual targets with C calibration. Its six-criterion family establishes KAHM margins over common features and raw-input RBF and establishes linear noninferiority. It does not establish the margins over selected positive-kernel departures, histogram gradient boosting, or a selected non-KAHM policy. Its KAHM nonlinear mean SMSE is 0.028912; the joint six-criterion requirement is not met.

A separate preprocessing-and-dimension study has 24 fresh repetitions and one established criterion out of four, namely linear noninferiority. Its primary nonlinear comparisons with independently selected positive-kernel reconstruction and a non-KAHM policy are unfavorable in mean loss. A residual-target computational reproduction regenerates the original 24 prescribed units with prior outcome knowledge and no new independent cohort; two of its four criteria are established. It cannot authenticate missing original execution records merely because its own calculations pass verification.

The fresh policy-augmentation study uses 24 repetitions and 192 tasks. Its base policy selects among ten non-KAHM families; one enlarged policy adds a centered KAHM correction and another adds a centered raw-history Nyström correction. Adding KAHM lowers nonlinear mean loss by 0.816% against the base policy and 0.658% against the added-RBF policy, but neither specified 3% margin is established. Both linear noninferiority criteria are met. Equal added search counts do not guarantee equal feature width or computation.

The natural-series policy evaluation uses 24 Weekly and 24 Hourly M4 training series, excludes pilot identifiers, and does not use official M4 Test data (M4 Competition, 2026). A/B/C/D are chronological blocks with separated targets. Its aggregate SMSE is 0.127612 for the base policy and 0.130512 after KAHM augmentation, an increase of 2.273%. Only three decisions change: one improves terminal loss and two worsen it. Shared unknown series sources preclude treating all series as independent source replications, and no independence-based confidence interval is assigned. This descriptive result is not a test of the central 26-procedure design.

S5.5 Scope of the research archive

Supervised input-metric geometry, directly trained reconstruction parameters, scaling changes, and online adaptation remain separate development investigations. Their targets, reference updates, and numerical conventions differ from the score-based direct-readout experiment. They are not treated as evidence that learned KAHM soft coordinates have Markov sufficiency or exact Koopman closure. The research index preserves their source locations, limitations, unsuccessful configurations, and study-specific records without pooling their observations. The separation between fixed-abstraction certification (Kumar & Kargaran, 2026) and external-target representation evaluation is maintained throughout.

S6 Reproducibility specification

The central experiment is identified by protocol `KAHM-SCORE-20261102-v1`. Its configuration, 256-task roster, primary family, and scientific-source manifest are supplied with the code. The preproduction source and the completed result release have separate identities. Source files, analysis records, and stage receipts are preserved without changing their historical contents.

S6.1 Four distinct reproduction tasks

A document build uses the supplied LaTeX sources, bibliography, and static tables. The table builder regenerates displays from compact released CSV records. Neither operation refits a forecasting model. A saved-loss checker independently recomputes all 2,560 paired differences, 40 intervals and decisions, and 416 C/D procedure-summary rows, conditional on the supplied losses.

Selected-model replay additionally requires prepared episodes and portable JSON/NPZ states. The full computational audit requires all candidate fits and retained predictions. The release inventory describes sixteen numerical archives totaling 9.024 GiB. The complete archives are not included in the compact materials, and a public download location has not been established. The compact records therefore support verification of saved-loss statistics but do not, by themselves, support replay of the released fitted models. The inventory identifies required bytes and membership; it does not establish their availability. Source sufficient to regenerate the cohort is supplied separately in the attributed research package, subject to the recorded environment requirements below.

Full regeneration uses the immutable scientific source in a new output tree. The command sequence is initialization, A/B preparation, fitting, complete fit sealing, C/D preparation, prediction, complete prediction sealing, scoring, analysis, audit, and release. Existing completed evidence is not an output workspace. Re-executing the same cohort is computational reproduction rather than additional independent statistical evidence. Timestamps and outer archive bytes can differ; substantive numerical arrays use the declared tolerances.

S6.2 Environment and portability

The recorded environment uses Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, threadpoolctl 3.6.0, and pytest 9.0.2. Production uses three fresh-process workers and one numerical thread per worker. The environment record also identifies the Linux platform, numerical libraries, and architecture. The preflight rejects a changed environment signature; source availability alone does not guarantee bitwise reproduction on another platform. A port requires explicit validation and a separately recorded environment or source amendment, not bypassing the check while claiming exact fidelity.

S6.3 Evidence provenance

The final release receipt binds the passing audit and released analysis. The source freeze records the prospective method; the earlier analysis receipt predates the completed audit. These records describe different points in the execution and are interpreted using their own scope. Hash checks establish byte identity, not independent scientific validity. The full numerical audit retains its shared-encoder limitation.

The accompanying repository documentation provides exact commands, archive restoration checks, source identities, and mappings from reported tables to input records. Related studies keep their own cohorts, confidence families, and availability statements.

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